Worksheet 5.4 — Curve Sketching with Derivatives

Mathematics: Analysis & Approaches SL (IB) · Unit 5: Differential Calculus

The Big Ideas

Critical points, the second derivative test, and inflection points reveal a curve's whole shape.

- Critical points: $f'(x) = 0$; classify via a sign change (first derivative test) or f'''s sign (second derivative test).
- $f'(c) = 0$ does not always mean an extremum — check for an actual sign change.

Worked Examples

Example 1 — First derivative test

Find and classify the critical points of $f(x) = x^3 - 12x$.

Solution. $f'(x) = 3x^2 - 12 = 3(x-2)(x+2)$; critical at $x = \pm 2$. Sign: $+, -, +$ across $-2, 2$.

Conclusion: local max at $x = -2$, local min at $x = 2$.

Example 2 — Second derivative test

Confirm the classification using f''.

Solution. $f''(x) = 6x$; $f''(-2) = -12 < 0$ (max), $f''(2) = 12 > 0$ (min).

Conclusion: confirmed.

Example 3 — Inflection point

Find the inflection point of $f(x) = x^3 - 12x$.

Solution. $f''(x) = 6x = 0 \Rightarrow x = 0$; sign changes from $-$ to $+$. $f(0) = 0$.

Conclusion: $(0, 0)$.

Example 4 — Critical point that is not an extremum

Show that $f(x) = x^3 + 2$ has a critical point at $x = 0$ that is not an extremum.

Solution. $f'(x) = 3x^2 \geq 0$ for every x — no sign change at $x = 0$.

Conclusion: $x = 0$ is a horizontal-tangent inflection point, not a max or min.

Example 5 — Full analysis of a quartic

Find and classify all critical points of $f(x) = x^4 - 8x^2$.

Solution. $f'(x) = 4x^3 - 16x = 4x(x^2 - 4) = 4x(x-2)(x+2)$; critical at $x = 0, \pm 2$. Sign of f': $-, +, -, +$ across $-2, 0, 2$.

Conclusion: local min at $x = -2$, local max at $x = 0$, local min at $x = 2$.

Example 6 — Finding both inflection points of a quartic

Find the inflection points of $f(x) = x^4 - 8x^2$ (from Example 5).

Solution. $f''(x) = 12x^2 - 16 = 0 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \pm\sqrt{4/3}$.

Conclusion: inflection points at $x = \pm\dfrac{2}{\sqrt{3}} \approx \pm 1.155$.

Example 7 — Determining global extrema on a closed interval

Find the global maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$.

Solution. Critical points: $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$ (both in $[-2, 3]$). Values: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$.

Conclusion: global max 18 at $x = 3$; global min -2 at $x = -2$ and $x = 1$ (tied).

Example 8 — Curve sketching summary for a cubic

Summarize the shape of $f(x) = -x^3 + 3x$ using critical points and concavity.

Solution. $f'(x) = -3x^2 + 3 = 0 \Rightarrow x = \pm 1$; $f''(x) = -6x$: $f''(-1) = 6 > 0$ (min), $f''(1) = -6 < 0$ (max). Inflection at $f'' = 0 \Rightarrow x = 0$.

Conclusion: local min at $(-1, -2)$, local max at $(1, 2)$, inflection at $(0, 0)$; concave up on $(-\infty, 0)$, down on $(0, \infty)$.

Example 9 — Why every cubic $x^3 + px + q$ has exactly one inflection point

Explain why $f(x) = x^3 + px + q$ always has exactly one inflection point, for any constants p, q.

Solution. $f''(x) = 6x$, which is zero at exactly one place, $x = 0$, and changes sign there (negative to positive) regardless of p, q.

Conclusion: every such cubic has exactly one inflection point, always at $x = 0$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find and classify the critical points of $f(x) = x^3 - 6x^2$.

Question 2

Confirm your classification using the second derivative test.

Question 3

Find the inflection point of $f(x) = x^3 - 6x^2$.

Question 4 — Challenge

Find and classify the critical points of $f(x) = x^4 - 32x$.

Question 5

Show that $f(x) = x^5$ has a critical point at $x = 0$ that is not an extremum, using $f'(x) = 5x^4$.

Question 6 — Challenge

Find all critical points and the inflection point of $f(x) = 2x^3 - 9x^2 + 12x$, and state the intervals of concavity.

Question 7

Find and classify the critical points of $f(x) = x^4 - 18x^2$.

Question 8

Find the inflection points of $f(x) = x^4 - 18x^2$ (from the previous question).

Question 9

Find the global maximum and minimum of $f(x) = x^3 - 12x$ on $[-3, 4]$.

Question 10

Summarize the shape of $f(x) = x^3 - 3x^2$ using its critical points and concavity.

Question 11 — Challenge

Find the global max/min of $f(x) = x^4 - 4x^2$ on $[-1, 3]$.

Question 12 — Challenge

Explain why $f(x) = x^3 + px + q$ always has exactly one inflection point, regardless of p, q.

Question 13 — Challenge

Find and classify the critical points of $f(x) = 3x^5 - 5x^3$ (hint: factor out x^3 from f').

Question Recap & Answer Key

Question Recap

1. Find and classify the critical points of $f(x) = x^3 - 6x^2$.
2. Confirm your classification using the second derivative test.
3. Find the inflection point of $f(x) = x^3 - 6x^2$.
4. Find and classify the critical points of $f(x) = x^4 - 32x$.
5. Show that $f(x) = x^5$ has a critical point at $x = 0$ that is not an extremum, using $f'(x) = 5x^4$.
6. Find all critical points and the inflection point of $f(x) = 2x^3 - 9x^2 + 12x$, and state the intervals of concavity.
7. Find and classify the critical points of $f(x) = x^4 - 18x^2$.
8. Find the inflection points of $f(x) = x^4 - 18x^2$ (from the previous question).
9. Find the global maximum and minimum of $f(x) = x^3 - 12x$ on $[-3, 4]$.
10. Summarize the shape of $f(x) = x^3 - 3x^2$ using its critical points and concavity.
11. Find the global max/min of $f(x) = x^4 - 4x^2$ on $[-1, 3]$.
12. Explain why $f(x) = x^3 + px + q$ always has exactly one inflection point, regardless of p, q.
13. Find and classify the critical points of $f(x) = 3x^5 - 5x^3$ (hint: factor out x^3 from f').

✓ Answer Key

1. $f' = 3x^2 - 12x = 3x(x-4)$: max at $x = 0$, min at $x = 4$
2. $f'' = 6x - 12$; $f''(0) = -12 < 0$, $f''(4) = 12 > 0$: confirmed
3. $6x - 12 = 0 \Rightarrow x = 2$, $f(2) = 8 - 24 = -16$: $(2, -16)$
4. $f' = 4x^3 - 32 = 4(x^3 - 8) = 4(x-2)(x^2 + 2x + 4)$; only real root $x = 2$ (quadratic factor has no real roots); sign changes $-$ to $+$: local min at $x = 2$
5. $f'(x) = 5x^4 \geq 0$ everywhere, no sign change: $x = 0$ is a horizontal-tangent inflection, not an extremum
6. $f' = 6x^2 - 18x + 12 = 6(x-1)(x-2)$: max at $x = 1$, min at $x = 2$; $f'' = 12x - 18 = 0 \Rightarrow x = 1.5$ (inflection); concave down on $(-\infty, 1.5)$, up on $(1.5, \infty)$
7. $f' = 4x^3 - 36x = 4x(x^2 - 9) = 4x(x-3)(x+3)$: min at $x = -3$, max at $x = 0$, min at $x = 3$
8. $f'' = 12x^2 - 36 = 0 \Rightarrow x^2 = 3 \Rightarrow x = \pm\sqrt{3}$
9. $f' = 3x^2 - 12 = 0 \Rightarrow x = \pm 2$; values: $f(-3) = 9$, $f(-2) = 16$, $f(2) = -16$, $f(4) = 16$; global max 16 (at $x = -2$ and $x = 4$, tied), global min -16 at $x = 2$
10. $f' = 3x^2 - 6x = 3x(x-2)$: max at $(0, 0)$, min at $(2, -4)$; $f'' = 6x - 6 = 0 \Rightarrow x = 1$ (inflection at $(1, -2)$); concave down on $(-\infty, 1)$, up on $(1, \infty)$
11. $f' = 4x^3 - 8x = 4x(x^2 - 2) = 0 \Rightarrow x = 0, \pm\sqrt{2}$ (only $0, \sqrt{2}$ in $[-1, 3]$); values: $f(-1) = -3$, $f(0) = 0$, $f(\sqrt{2}) = -4$, $f(3) = 45$; global max 45 at $x = 3$, global min -4 at $x = \sqrt{2}$
12. $f''(x) = 6x$, zero only at $x = 0$ and changing sign there regardless of p, q — so exactly one inflection point always exists, at $x = 0$
13. $f' = 15x^4 - 15x^2 = 15x^2(x^2 - 1) = 15x^2(x-1)(x+1)$: critical at $x = 0, \pm 1$; sign of f': $+, -, +$ crossing -1 and 1 but $x = 0$ has no sign change (double root): max at $x = -1$, min at $x = 1$, $x = 0$ is not an extremum